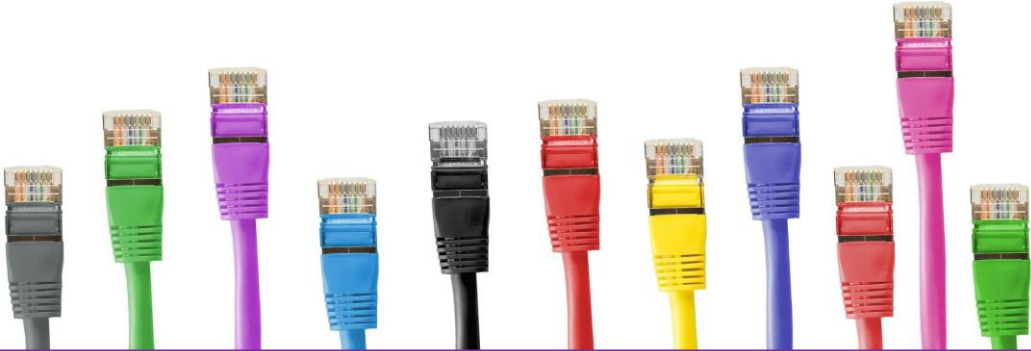




SOF203 Fundamentals of Network Technology

Revision

Xiamen University Malaysia



Final Exam

- 2 hours
- 4 Questions – with sub-questions
 - No true/false, no MCQ
- You can bring calculator
 - No smartphone/tablet/smart watch
- Draft papers are provided
 - Don't take your own notes/papers
- Plan your time
 - e.g., ~30 mins for Q1 & Q3, ~1hr for Q2

Course Learning Outcomes

Upon the completion of the course, the students are able to:

1. *Describe the theory and principles of computer network architecture and protocols, especially TCP/IP protocol suite.*
2. *Solve problems related to hardware and software components of computer networks.*
3. *Analyse the network flow on a local network.*
4. ~~*Perform planning and configuration for a simple network.*~~

1. *Describe the theory and principles of computer network architecture and protocols, especially TCP/IP protocol suite*

- Chapter 1

- Types of networks (coverage area)
- State the names of the 5 layers/ 7 layers model
- Explain/describe the responsibilities of each layer (5 layer model)
- Concept of encapsulation/decapsulation

- Chapter 2

- Explain TDM/FDM (with diagram)
- Explain simplex/half-duplex/full-duplex transmission (with diagram)
- Explain the differences between circuit switching and packet switching

2. Solve problems related to *hardware and software components* of computer networks

- Chapter 2

- Compute various delays – transmission delay, propagation delay, processing delay, etc.
- Explain the differences between the delays

- Chapter 3

- Analyse ARP packet traces (request/reply)
- Switch operation (switch table) – self learning, forwarding
- Describe active/passive scanning in WLAN

2. Solve problems related to *hardware and software components* of computer networks

- Chapter 4

- Subnetting
 - Identify network/broadcast/usable IP addresses based on a given address block
- Apply Dijkstra's algorithm to solve routing related problem
- Apply distance-vector algorithm to solve routing related problem

3. *Analyse the network flow* on a local network

- Chapter 3
 - Addressing- identify (source/destination) IP/MAC addresses of a packet
- Chapter 5
 - Identify sequence/acknowledgment numbers of TCP segments in
 - TCP connection management (3-way handshake, data transfer, connection teardown)
 - TCP reliable data transfer
- Chapter 6
 - Analyse HTTP (request/reply) messages